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PAPER_993: TXS 0506+056 Blazar Jet F_U_Bi_i Modulation

Abstract

We compute the F_{U,Bi_i} jet modulation curves for TXS 0506+056, the IceCube neutrino blazar ($M_{BH} = 3 \times 10^8 M_\odot$, $a = 0.95$, $B = 5000$ G). The peak jet modulation factor reaches $M_{jet} = 3.3\times$ at on-resonance Γ_0 .

1. Jet Modulation

$$M_{jet}(\Gamma_0) = 1 + A_{jet} \cdot e^0 = 1 + 2.3 = 3.3$$

with $A_{jet} = 2.3$ for TXS 0506+056 (higher than CenA due to extreme spin $a = 0.95$).

2. Neutrino Production

The $3.3\times$ jet power enhancement at SCm resonance provides a natural explanation for the IceCube neutrino excess: enhanced proton acceleration in the jet produces pions that decay to neutrinos.

3. F_U_Bi_i at Horizon

The buoyancy-gravity balance at the ergosphere boundary (r_H for $a = 0.95$) gives $F_{U,Bi_i} < 0$ (inward-dominant), with the jet modulation acting as a perturbative correction.

4. Implementation

File: fubi_inside_outside.py, class TXS0506FUBiCurves. CP4 class #577.